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# WORK

## *When you grow up, you might not have a job*

We have no idea what the job market will look like in 2050. It is generally agreed that machine learning and robotics will change almost every line of work – from producing yoghurt to teaching yoga. However, there are conflicting views about the nature of the change and its imminence. Some believe that within a mere decade or two, billions of people will become economically redundant. Others maintain that even in the long run automation will keep generating new jobs and greater prosperity for all.

So are we on a verge of a terrifying upheaval, or are such forecasts yet another example of ill-founded Luddite hysteria? It is hard to say. Fears that automation will create massive unemployment go back to the nineteenth century, and so far they have never materialised. Since the beginning of the Industrial Revolution, for every job lost to a machine at least one new job was created, and the average standard of living has increased dramatically.[1] Yet there are good reasons to think that this time it is different, and that machine learning will be a real game changer.

Humans have two types of abilities – physical and cognitive. In the past, machines competed with humans mainly in raw physical abilities, while humans retained an immense edge over machines in cognition. Hence as manual jobs in agriculture and industry were automated, new service jobs emerged that required the kind of cognitive skills only humans possessed: learning, analysing, communicating and above all understanding human emotions. However, AI is now beginning to outperform humans in more and more of these skills, including in the understanding of human emotions.[2] We don't know of any third field of activity – beyond the physical and the cognitive – where humans will always retain a secure edge.

It is crucial to realise that the AI revolution is not just about computers getting faster and smarter. It is fuelled by breakthroughs in the life sciences and the social sciences as well. The better we understand the biochemical mechanisms that underpin human emotions, desires and choices, the better computers can become in analysing human behaviour, predicting human decisions, and replacing human drivers, bankers and lawyers.